NETWORK SECURITY MONITORING ARCHITECTURE LAB FOR THREAT DETECTION AND ANALYSIS

Security Operation Center (SOC) Analysis Document

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Project Type:

Security Monitoring and Network Defense Lab (SOC Simulation Environment)

Objective:

Design and implement a simulated SOC environment to collect and analyze security events across windows and Linux systems for detecting and investigating malicious activity

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EXECUTION SUMMARY

This document outlines a virtualized security monitoring environment designed to simulate a segmented enterprise network with centralized detection and logging capabilities.

The architecture demonstrates how firewall enforcement, intrusion detection and SIEM-based correlation operate together to provide end to end security visibility

This lab is implemented using:

- Pfsense for network segmentation and traffic enforcement
- Snort for intrusion detection and network inspection
- Splunk for centralized logging, correlation and analysis
- Virtualize for isolated infrastructure simulation
- Kali Linux
- Windows 10
- Windows sever 2025
- Virtual Box for isolated infrastructure simulation

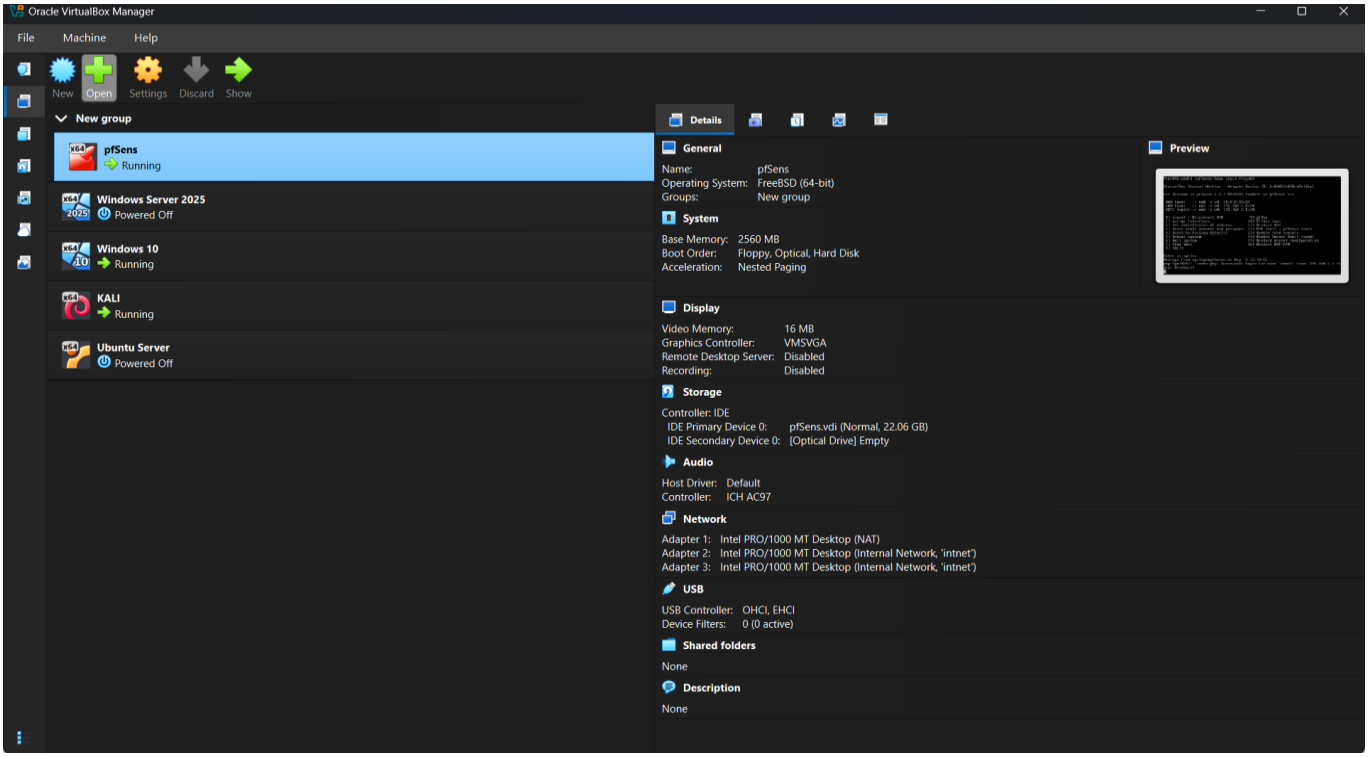

Figure 1: Main Architecture

IDENTIFY

Objective

Establish system boundary, assets and trust zones

Implementation

- Network segmentation defined as:
  - WAN – external simulation zone
  - OPT1 – untrusted network segmentation
  - LAN – trusted internal network
- Pfsense serves as the central routing and policy enforcement layer
- Critical assets include:
  - Security monitoring stack (Stock, Splunk)

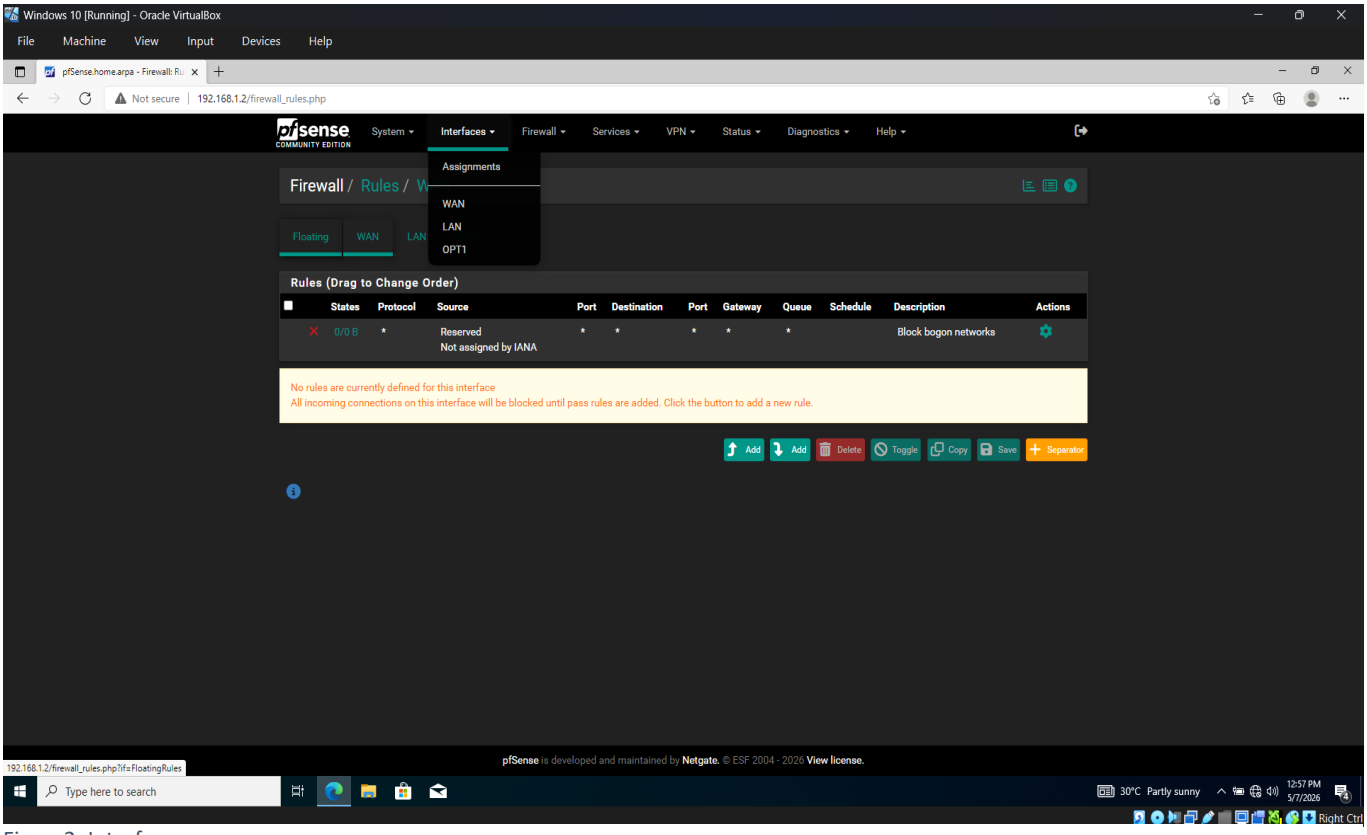

Figure2: Interfaces

Outcome

Clear definition of trust boundaries and asset visibility to support security control designed

PROTECT

Objective

Enforcement access control and limit unauthorized communication

Implementation

- Pfsense enforces inter-zone traffic control policies
- Firewall rules regulate communication between WAN, OPT1 and LAN

- Attack surface reduced through network isolation
- Separation of attacker simulation environment from internal assets

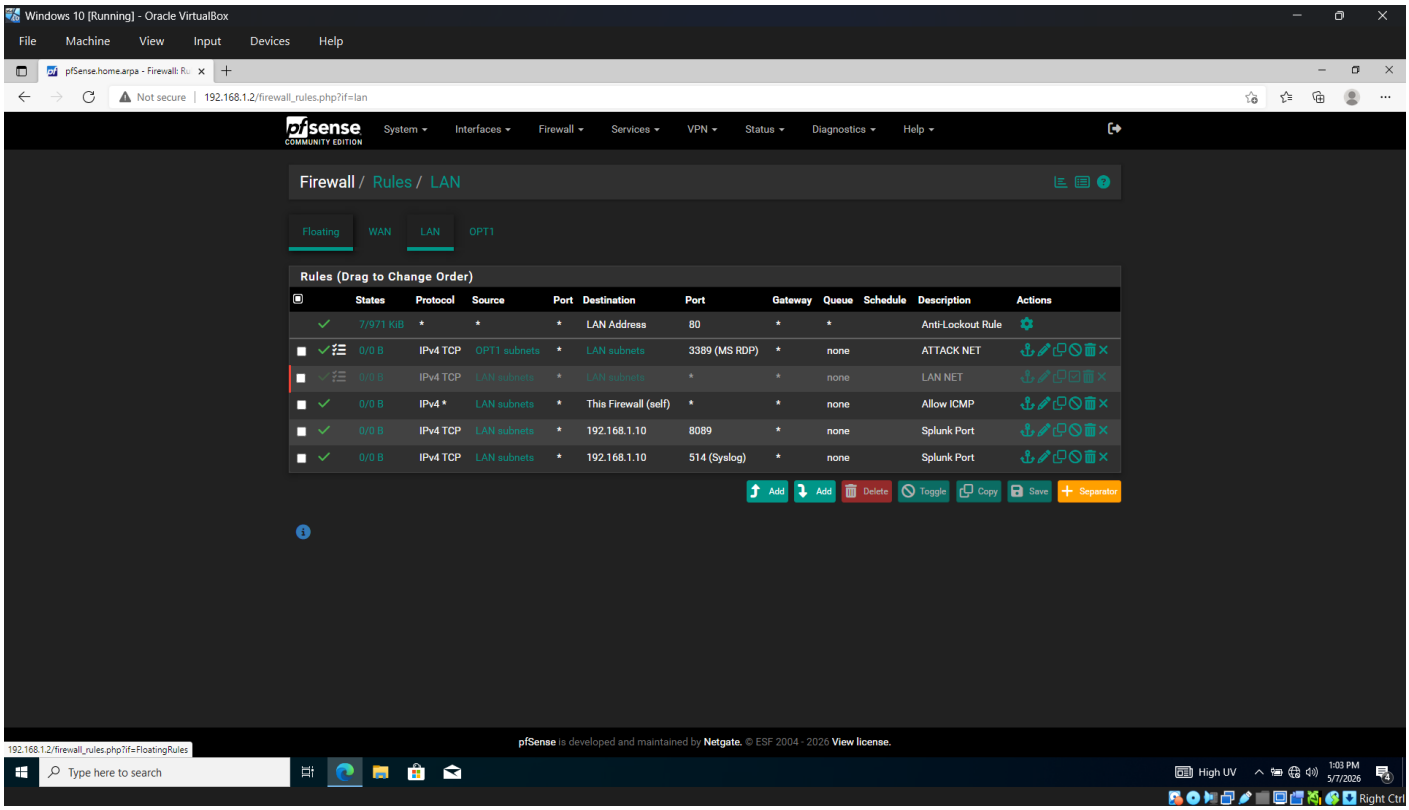

Figure 3: Firewall Rules

Outcome

Controlled and restricted communication paths across network zones, enforcing segmentation and least privilege

DETECT

Objective

Identify malicious or anomalous network activity.

Implementation

- Snort deployed for real-time intrusion detection
- Signature-based and rule-based traffic analysis
- Detection capabilities include:
- Port scanning and reconnaissance behavior
- Brute-force authentication attempts
- Suspicious connection patterns

- Security events forwarded for centralized logging

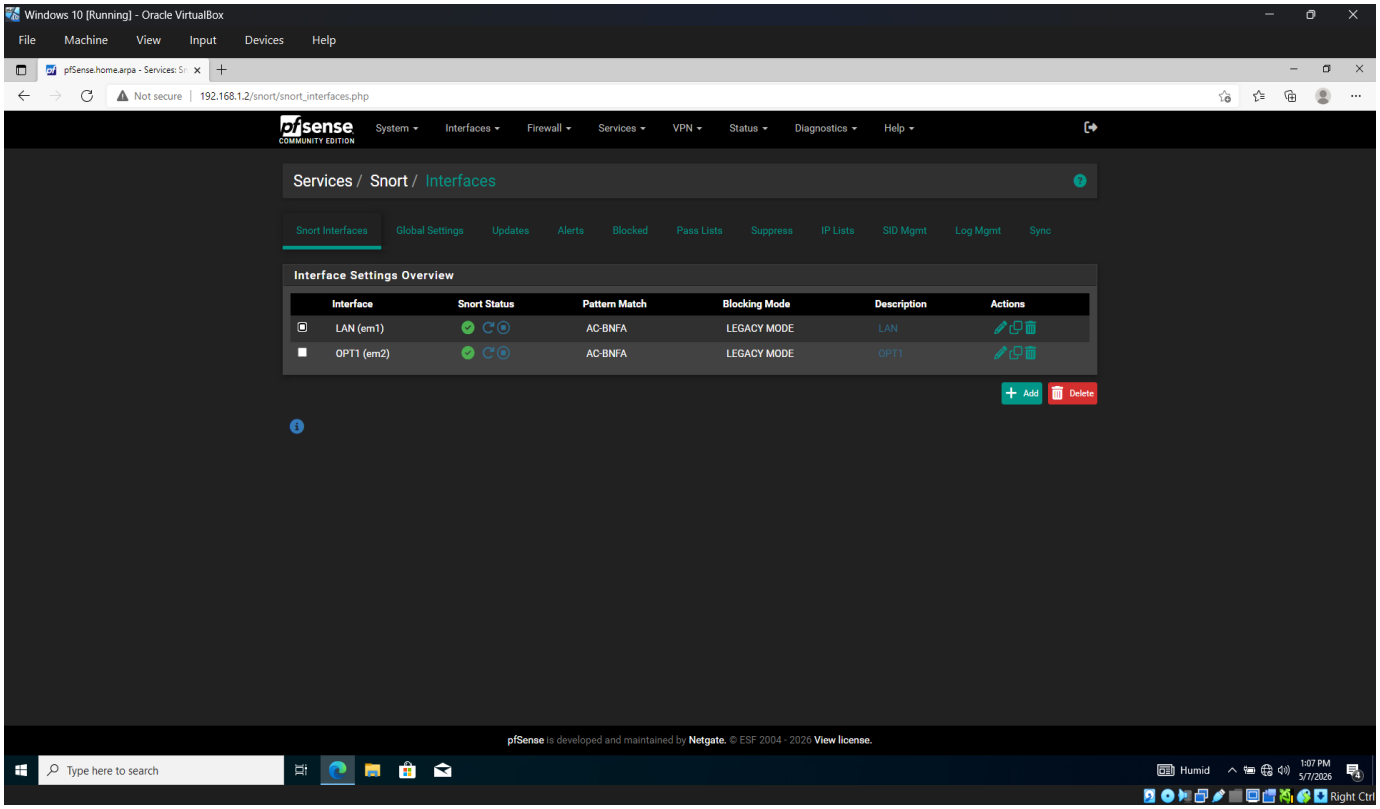

Figure 4: Snort Interfaces

Outcome

Network-based threats are detected and logged at the inspection layer with alert generation.

RESPOND

Objective

Enable investigation and analysis of security events.

Implementation

- All logs centralized in Splunk
- Correlation of multi-source events (firewall, IDS, host logs)
- Enables:
  - IP behavior analysis
  - Attack timeline reconstruction
  - Alert aggregation across systems

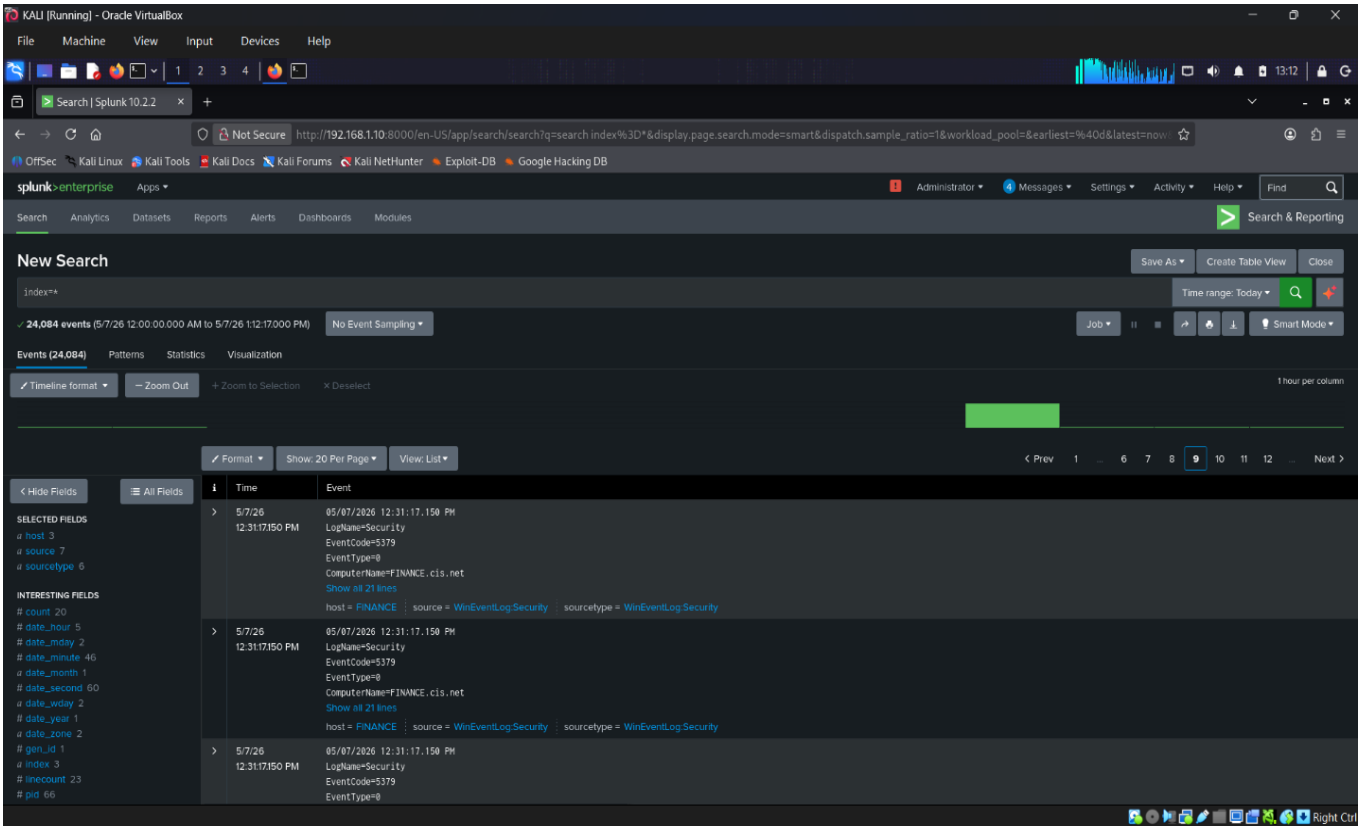

Figure 5

Outcome

Improved incident analysis through centralized visibility and correlation of security telemetry.

RECOVER

Objective

Ensure system recoverability and operational continuity.

Implementation

- Virtual Box snapshots enable full system rollback
- Environment isolation ensures no production impact
- Systems can be restored to a clean baseline after testing

Outcome

Repeatable and recoverable lab environment supporting continuous experimentation.

ARCHITECTURE OVERVIEW

The system implements a layered security monitoring model:

- Network Control Layer: pfSense (segmentation and enforcement)
- Detection Layer: Snort (packet inspection and alerting)
- Analytics Layer: Splunk (SIEM correlation and analysis)
- Infrastructure Layer: Virtual Box (isolated simulation environment)

This structure reflects a simplified Security Operations Center (SOC) architecture.

OBSERVABILITY MODEL

Security visibility is achieved through multiple telemetry sources:

- Firewall logs (traffic enforcement layer)
- IDS alerts (network inspection layer)
- SIEM correlation (central analytics layer)

This multi-layered approach enables detection coverage across different points in the attack lifecycle.

DESIGN CONSTRAINTS

The environment has the following limitations:

- Lack of enterprise-grade SPAN or network tap functionality
- IDS visibility dependent on network placement and routing design
- Virtual traffic does not reflect real production-scale volume or diversity
- Limited hardware-level telemetry due to virtualization constraints

These constraints are inherent to lab-based simulation environments

SECURITY PRINCIPLES IMPLEMNTED

- Network segmentation and trust zoning
- Defense-in-depth architecture
- Centralized logging and correlation
- Separation of enforcement, detection, and analysis layers
- Least privilege network access design

CONCLUSION

This lab demonstrates a structured security monitoring architecture in a virtual environment.

It models key enterprise SOC concepts, including network segmentation, intrusion detection, and centralized security analytics.

The design provides a foundational framework for understanding how security controls integrate across multiple layers to support detection, analysis, and response workflows in modern cybersecurity operations.

LESSON LEARNED

- Placement matters more than tools: IDS effectiveness depends on correct network positioning; poor placement creates blind spots regardless of configuration quality.

- SIEM visibility is not automatic: Log ingestion must be deliberately designed (forwarders, indexes, and parsing). If not, data exists but is invisible.

- Single data sources are insufficient: Firewall logs alone cannot explain incidents. Effective investigation requires correlation across firewall, IDS, and SIEM data.

- Correlation is the real skill: Security monitoring is not about tools, it’s about connecting signals across multiple layers to reconstruct events accurately.